

Comparative Analysis: ChatGPT vs. Healthcare Professionals in Addressing Type-1 Diabetes FAQs in English and Arabic

Mohammad D. Alahmadi *

Mohammed Al-Sofiani

Sarah Albunyan

Abdullah Alahmadi *

Osama Alzobaidi *

*Department of Software Engineering,
University of Jeddah, Jeddah, Saudi Arabia



This work was funded by the
University of Jeddah, Jeddah,
Saudi Arabia, under grant No.
(UJ-23-FR-78).

Abstract

Patients with Type-1 Diabetes (PWT1D) often need immediate answers to critical questions related to hyperglycemia, hypoglycemia, diagnosis, and insulin management. Access to specialists like endocrinologists may be limited, leading many patients to seek help from social media, where responses may not always be accurate or timely. ChatGPT, a leading AI language model, has the potential to assist PWT1D by understanding prompt and providing accurate answers in both English and Arabic. This study aims to compare the quality of responses from ChatGPT with those from healthcare professionals to evaluate its effectiveness in addressing PWT1D-related FAQs.

Methodology

For this study, a survey was conducted among endocrinologists, diabetes educators, and dietitians to gather frequently asked questions (FAQs) about Type-1 Diabetes (T1D) in English and Arabic. After open coding and categorizing these questions, two questions per category were selected without duplication. Using zero-shot prompting, ChatGPT-3.5 generated answers for the FAQs. A diverse panel of experts and individuals with T1D evaluated the responses for accuracy and empathy using a Likert scale, comparing ChatGPT's responses to human-generated ones, assessing preference, and distinguishing AI from human responses.

Results

A total of 66 FAQ pairs were collected from endocrinologists, diabetes educators, and dietitians, with an equal distribution of 33 questions in both English and Arabic. These FAQs were categorized into seven key themes: hyperglycemia, hypoglycemia, treatment, management, diagnosis, insulin, and diet.

Figures 1 and 2 summarize the accuracy & empathy ratings of responses generated by ChatGPT versus healthcare professionals, with evaluations done by eight specialists. Notably, ChatGPT performed similarly to human experts in terms of accuracy, especially in Arabic, where both had identical scores. For English responses, ChatGPT slightly outperformed human responses. Empathy ratings were comparable between ChatGPT and human answers, with ChatGPT showing higher empathy in English answers.

Overall, healthcare professionals and patients showed a slight preference for ChatGPT-generated responses (53.47%) compared to human-generated ones (46.53%), indicating the potential of AI tools in diabetes management.

Conclusions

This study shows that ChatGPT can deliver accurate and empathetic responses to frequently asked questions from patients with Type-1 Diabetes (PWT1D) in both English and Arabic. Its performance was comparable to healthcare professionals, indicating its potential in diabetes management and patient support. Future research should focus on enhancing ChatGPT's performance and exploring other AI models for improved patient care.

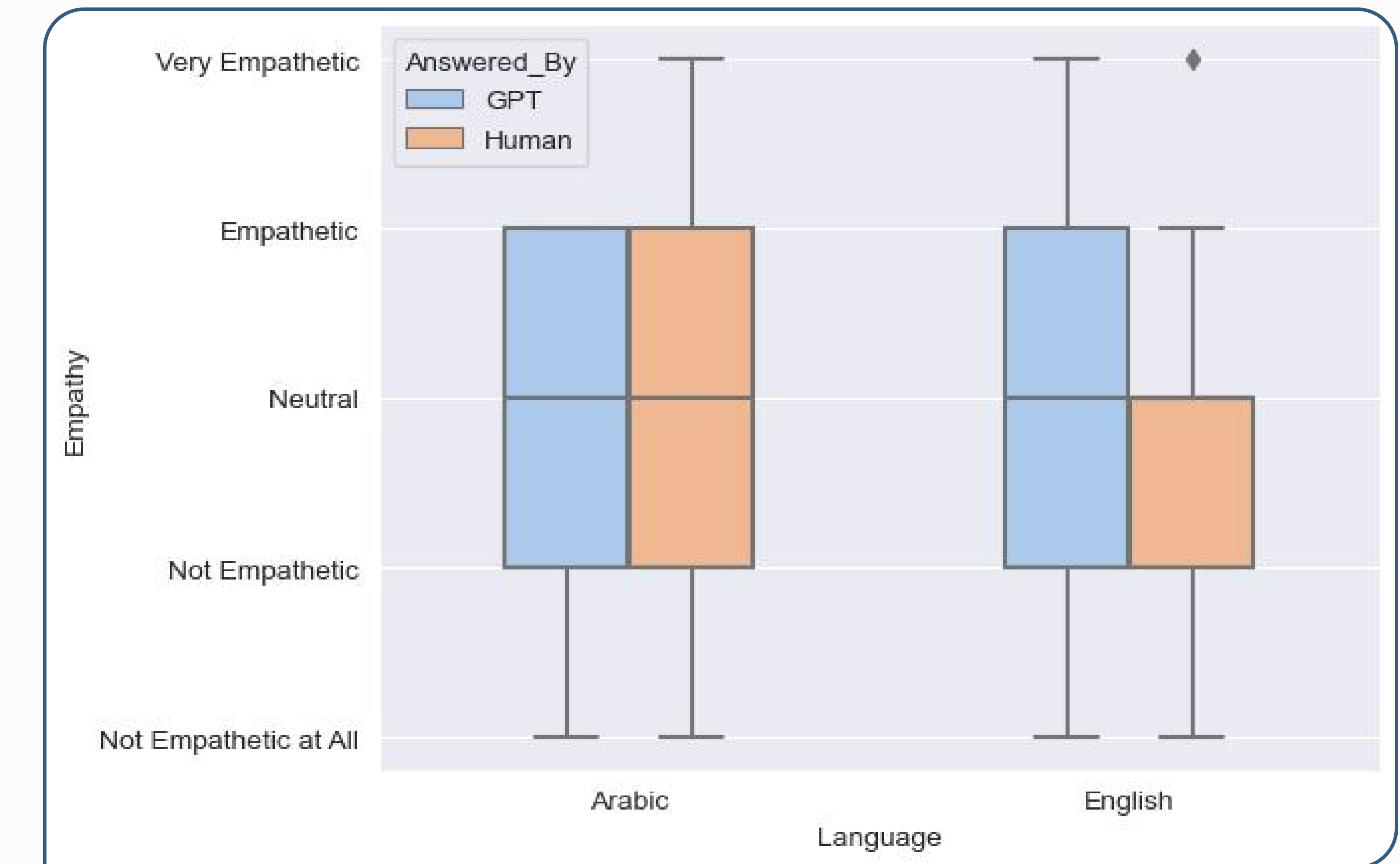


Figure 1

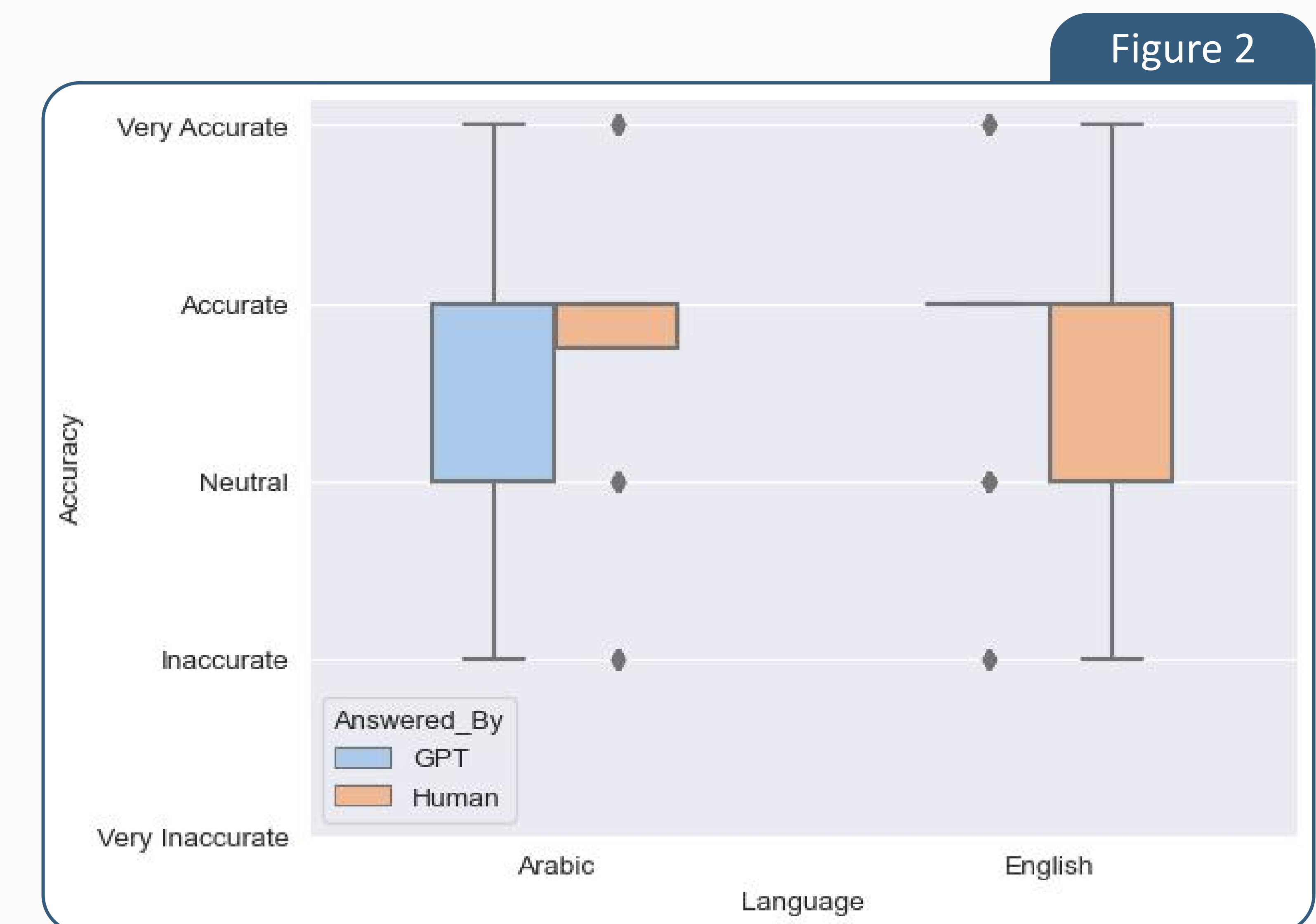


Figure 2